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Executive Summary

The British Isles host a wide range of copper occurrences in diverse geological settings. Major historic and prospective copper provinces include the Late-Palaeozoic “vein-style” orebodies of SW England (Cornwall–Devon), Volcanogenic Massive Sulphide (VMS) deposits in Wales (notably **Parys Mountain**, Anglesey) and NW Scotland (Lewisian); porphyry-style copper in Wales and Scotland (e.g. Coed-y-Brenin, North Wales; Llandeloy, SW Wales); sediment-hosted red-bed copper in England (e.g. Cheshire); and Dalradian stratiform copper in Scotland (e.g. Vidlin Ness, Shetlands)【64†L200-L208】【66†L302-L310】. Parys Mountain (Anglesey) is a world-class polymetallic VMS, historically Britain’s largest copper mine and now being redeveloped; its 2023 resource is ~16.1 Mt @ 0.98% Cu (~158 kt Cu)【40†L55-L64】. Other high-potential onshore targets include Coed-y-Brenin (Snowdonia NP, ~200 Mt @ 0.3% Cu)【64†L239-L247】, the Llandeloy porphyry system (SW Wales), and the Alderley Edge red-bed copper (Cheshire) with historic grades ~2.1% Cu【66†L314-L319】. Known occurrences in Ireland (e.g. Allihies, Avoca, and certain Irish-type Zn–Pb deposits with Cu) add further targets, though none currently produce.

In UK/Irish offshore waters, potential copper is associated with ferromanganese crusts on Atlantic seamounts and polymetallic nodules on deep abyssal plains – both of which contain copper along with Co–Ni–Mn【93†L198-L204】【93†L206-L214】 – and rare seafloor massive sulfide systems at hydrothermal vents (mid-Atlantic Ridge, Rockall Bank, etc.). No commercial seabed mining exists to date; any new offshore extraction would require a marine licence under the UK Marine & Coastal Access Act 2009【102†L64-L72】 (or Irish foreshore licences) and must navigate strict environmental protections (MPAs, MCZs, Natura 2000, OSPAR, depth limits).

This report examines host-rock types, deposit styles and controls; reviews historic and current copper mines (locations, production, grades, owners, status); surveys known occurrences and recent exploration; outlines offshore geology and target types; and identifies priority target areas. Constraints (environmental, legal, technical) are summarized. Where available, data are compiled into tables (coordinates, geology, prospectivity, evidence, access, infrastructure, sensitivity) and illustrated with maps and timelines. Key sources include BGS (“Minerals in Britain” volumes, national datasets), Geological Survey Ireland, peer-reviewed papers and industry/government reports.

Geology of Copper Mineralization in the British Isles

Copper in Britain and Ireland occurs in several deposit types, reflecting the complex geology:

- **Vein-style (orogenic/hydrothermal) Cu** – Associated mainly with *Hercynian* (Late Carboniferous–Permian) granites and volcanic belts. Classic examples are Cornwall–Devon vein systems (Cornish lodes), where Cu (with Sn, W, Ag) occurs in quartz–baryte–fluorite veins. These yielded >2 Mt Cu historically【66†L334-L343】. Similar vein copper was mined in Anglesey and North Wales (e.g. at Parys Mountain, Nant y Moch) and in parts of Ireland (e.g. Allihies, Co. Cork). Structural controls are steeply plunging lodes in shear zones related to granite emplacement. *Note:* Historical grades were often 1–3% Cu (in high-grade veins), but averaged much lower (e.g. 0.3–0.5% over larger stopes).

- **Volcanogenic Massive Sulphide (VMS)** – Cu occurs in stratabound volcanic-related sulphide deposits. The prime British example is **Parys Mountain** (North Wales, Anglesey): an extensive Ordovician VMS containing Cu–Zn–Pb–Ag±Au. BGS reports that Parys was mined for Cu (and Pb–Zn) since the Bronze Age, and its 1997 resources were ~6.5 Mt @ 2.3% Cu, 5.3% Zn, 2.7% Pb (with 39 g/t Ag)【64†L200-L208】. Anglesey Mining plc (current owner) has since expanded this to ~16.1 Mt @ 0.98% Cu【40†L55-L64】. Structurally, the sulphides are lenses associated with silicified volcanic exhalites. A second VMS style occurs in Precambrian Lewisian meta-volcanics (NW Scotland): the *Besshi-type* **Kerry Road** deposit (Gairloch area) is a Cu–Zn–Au VMS lens (~1% Cu)【9†L83-L88】 formed in a sheared banded-iron formation. Dalradian stratiform Cu sulphides also occur in Scotland (e.g. Vidlin Ness, Shetlands, with up to 1.19% Cu over 10 m)【64†L172-L180】.

- **Porphyry/Cu-Au** – Several **porphyry-style Cu** occurrences have been identified (typically Caledonian age), usually sub-economic disseminated Cu±Au in felsic intrusives. Notable examples: Coed-y-Brenin (North Wales) – ~200 Mt @ 0.3% Cu【64†L239-L247】 within a volcanic-intrusive complex, and Llandeloy (SW Wales) – a concealed Cu–Au system in Proterozoic rocks identified by drilling【64†L239-L247】. In Scotland, *epithermal/porphyry* alteration is seen at Lagalochan (Argyll) and minor porphyry-associated Cu at Fore Burn, Black Stockarton (Southern Uplands)【66†L268-L276】. These deposits are controlled by intrusive centers (often unexposed) and late Caledonian structures.

- **Sediment-hosted (Red-bed) Cu (SCD)** – Triassic fluviatile sandstones in central England host Cu–Co (with Pb–Zn etc.) as continental "red-bed" deposits. The classic Cheshire deposits (Alderley Edge–Macclesfield) occur in the Sherwood Sandstone, where copper carbonates and oxides accumulated in permeable channels. Worked ore from Alderley averaged ~2.1% Cu【66†L314-L319】. Mineralization is stratigraphically controlled by channel sands and fault traps beneath Mercia Mudstone cover. Similar Devonian–Carboniferous sandstone-hosted Cu occurs sporadically in Scotland (Old Red Sandstone areas【66†L328-L334】).

- **Skarn/Contact** – Cu is also present in skarn zones around granites. For instance, at Ecton Hill (Staffordshire) Cu–Zn–Pb–Ag skarn veins were mined historically (mostly for tungsten), and Cu mineralization occurs at contacts of Cornubian granites (e.g. North Cornwall). Dalradian carbonate-hosted Cu (e.g. Cae Coch, North Wales) is essentially a skarn/massive pyrite deposit.

Each deposit type has characteristic controls – for example, Cornish vein copper followed steep faults off Carnmenellis granites; Parys VMS follows stratigraphy of submarine lavas; Alderley Channelites follow paleo-river beds and faults. These geological templates define the known occurrences and guide exploration (see Figure 1 map of deposit types and sample sites below).

Historic and Current Copper Mines

- **Parys Mountain (Anglesey, North Wales):** *Polymetallic VMS.* The largest copper mine in Britain's history. Bronze–Iron Age exploitation is recorded; intensive mining in the 1760s–90s made Parys "the world's largest copper producer" at the time【25†L95-L104】. Over 100 kt Cu was produced (with Zn–Pb–Ag) by the late 18th century. Production halted when overseas Cu undercut prices (post-1790), but Parys resumed in 19th century on a smaller scale. **Current status:** Anglesey Mining plc holds planning permission and aims to reopen it. The site has existing infrastructure (headframe, 300 m shaft) and supportive local authority【39†L55-L60】. Its *JORC*-compliant resources (2023) are ~16.1 Mt @ 0.98% Cu (158 kt Cu) plus Zn–Pb–Ag【40†L55-L64】.

- **North Wales Historic Vein Mines (e.g. Clwydd, Bwlch y Ddwyfor, Nant-y-Moch):** Copper-bearing quartz veins were mined on the Harlech Dome and Snowdon area. Production was modest (few 1000s t Cu) and mostly ceased by early 20th century.

- **South-west England (Cornwall–Devon):** *Vein copper.* From the 17th–19th centuries hundreds of lode mines yielded copper along with tin. In aggregate, SW England produced well over 2 Mt of copper【66†L334-L343】 (mostly as by-product of tin mining). Major centres included *Dolcoath, Herland,* and mines around Redruth–Camborne. Peak production was late 18th century (Royal Navy demand), dropping by 1860s due to foreign competition【25†L95-L104】. **Current status:** All Cornish copper mines are closed. Remediation and heritage preservation (Cornish Mining WHS) are active, but no production.

- **Central England (Alderley Edge, Cheshire):** *Sediment-hosted Cu.* Numerous small mines (1619–20th c) exploited Triassic copper ores. Output was tens of thousands of tonnes of high-grade oxide copper (avg ~2.1% Cu)【66†L314-L319】. Production ceased by early 20th c as ore depleted. Remnants of workings exist near Congleton.

- **Scotland (NW Highlands and Islands):** Small veins and stratiform Cu. *Vidlin Ness* (Shetland) – exploration drilling (1970s) found a stratabound lens ~10 m thick at 1.19% Cu【64†L172-L180】. *Glenelg–Gairloch (Kerry Road):* the Kerry Road VMS (1.0% Cu) was delineated by drilling but proved sub-economic【9†L83-L88】. *Lagalochan (Argyll):* Cu–Au veins near granites, explored by drilling (Late 20th c). *Ballachulish, Cairngarroch, Fore Burn (Highlands):* minor porphyry/skarn-style Cu ± Mo, identified by BGS reconnaissance【66†L268-L276】. None of these have been mined commercially; they remain prospects (often peripheral interest in Ni or Au).

- **Ireland:** Historical copper mining occurred at *Allihies (Co. Cork)* and *Avoca (Co. Wicklow)* in Paleozoic volcanic settings (Cu–Au–Ag–Sb ores). These Victorian-era mines produced many 10s of thousands of tonnes of copper. In recent years, exploration companies (e.g. Conroy Gold, Galantas) have targeted porphyry/vein Cu–Au systems (Clontibret, Galmoy) and Irish-type Zn–Pb(±Cu) deposits (Stonepark, Pallas Green). Some base-metal mines (Silvermines, Lisheen) contained Cu as by-product (Silvermines had Cu–Co–Ni associated with Zn–Pb ore). **Current status:** No operating copper mines; sporadic exploration with modern geophysics/drilling.

Known Occurrences and Exploration Results

Modern exploration and surveys have identified many copper prospects across Britain and Ireland:

- **North Wales/Snowdonia:** Besides Parys, exploration by MEIGA/MRP in the 1980s–90s found *Coed-y-Brenin* (near Dolgellau): a *porphyry Cu* system (~200 Mt @ 0.3% Cu)【64†L239-L247】. Smaller shear-hosted Cu values were noted at *Maeshafn*, *Brynheulog*, and elsewhere (up to ~1% over metres). Glencore-Kinross drilled *Llandinam* (Central Wales) for porphyry Cu–Au with some low-grade Cu zones. Prospecting in Snowdonia has also targeted strata-bound Cu (e.g. Cae Coch massive pyrite of Ordovician age【64†L200-L208】).

- **South-west England:** BGS's MRP intersected *low-grade disseminated Cu* near Higher Coombe (Devon)【64†L187-L196】. Mineralised veins and stockworks were found in the Carboniferous–Devonian succession (±Sedex style). No economic deposit emerged, but several targets (Coombe, Dartmoor fringe) have been postulated. There is renewed interest in Cornwall: the North Crofty tin-copper project now targets residual copper in old tailings; see also Redmoor Cu–Sn project (recent drilling).

- **England (Midlands/Lakeland):** Exploration around Alderley Edge used drilling and geophysics; deep targets under cover remain of interest. In the Lake District, historic copper mining at *Ecton* (Staffs) and *Park Mill Mine* (Lancs) exemplified carbonate-hosted Cu. BGS mapping notes anomalous Cu in Borrowdale volcanics, but no large deposits.

- **Scotland:** BGS studies (MRP reports) identified stratiform copper in the Dalradian (Shetland, Loch Maree), and in Dalradian and Torridonian (NW Scotland) possibly similar to Kerry Road 【37†L49-L58】【64†L172-L180】. The Anorthosite Complex (Laggan, Fore Burn) shows elevated Cu values (unexploited). Recent compilations highlight the *Knock intrusion* (Aberdeenshire): Ni–Cu–PGE exploration (3 Mt at 0.27% Cu)【66†L288-L297】 but copper is subordinate to Ni.

- **Ireland:** Recent GSI surveys have identified new target areas: Carboniferous basins for Irish-type Cu–Zn–Pb (e.g. Stonepark Zn–Cu prospect), porphyry systems (Orlock Bridge Fault targets in Monaghan–Armagh), and "Oxford co-type" carbonate Cu (Gortdrum deposit: high-grade Zn–Pb with Cu–Ag and As–Sb【70†L10-L14】). Allihies was drilled by Placer (1980s) revealing remaining Cu anomalies (IGP site). Some Zn–Pb exploration holes (e.g. Tara Deep) also intersected Cu-rich veins, showing the presence of deeper Cu systems. Overall, Ireland has under-explored Cu potential, especially where old mines existed, but systematic data are patchy.

Offshore Seabed Geology and Mineralization

The UK and Irish Continental Shelves are mostly covered by sediments, but key geological features and analogue deposits suggest offshore copper potential:

- **Ferromanganese Crusts:** In areas of significant volcanic relief (seamounts, ridges) at depths ~400–4,500 m, cobalt-rich ferromanganese crusts grow on hard substrate【93†L198-L204】. These crusts contain Co, Ni, Mn and also Cu (typically 0.3–1.0%) along with platinum-group elements【93†L206-L214】. Examples in the Atlantic include seamounts west of Ireland and the Rockall Bank (e.g. Anton Dohrn, Rosemary Bank); UK/EU contractors (e.g. Cornwall Metals) have reported thick Co-crusts in international seabed surveys. These crusts are within EEZs, so extraction would be under national jurisdiction.

- **Polymetallic Nodules:** Abyssal plains (3–6 km deep) worldwide host manganese nodules ("manganese nodules") enriched in Mn, Ni, Cu, Co. While the Pacific (Clarion–Clipperton) is richest, Atlantic plains (e.g. covered by UK/Irish data across Rockall Trough, Hatton Bank) contain nodules at lower concentrations. No UK company has active claims, but ISA contracts exist globally. The nodules grade roughly 1–2% Cu (depending on location) but require deep-sea mining technology.

- **Seafloor Massive Sulphides:** Hydrothermal vents at mid-ocean ridges or Atlantic rift zones form Cu–Zn–Au–Ag massive-sulphide deposits ("black smokers"). The Mid-Atlantic Ridge (beyond UKCS) and hidden vent sites (e.g. attempted target at Irish Newbarnes et al.) are potential sources of copper-rich sulphides. Within UK/Ireland waters, the Faroe–Shetland Trough and continental margin fault zones (e.g. Hebridean Basin) might host buried VMS-like deposits, but none are confirmed by drilling. **Note:** No active offshore mining exists in UK waters; any exploitation would face strict environmental assessment (Habitats Directive, OSPAR) and technological challenges (depth, remoteness).

- **Placer Deposits:** Copper placers (native copper in sands) are virtually non-existent around Britain's shores. Ancient submarine landslides or turbidites could conceivably concentrate heavy minerals, but no significant copper placer targets are known.

High-Priority Target Areas

Based on geology and past data, we highlight these priority areas (coordinates approximate):

- **Parys Mountain, Anglesey (53.38°N, –4.35°W):** Known VMS orebody; near-surface mineralization; positive feasibility studies【39†L55-L60】. Excellent infrastructure (port at Amlwch, grid power) and planning already granted【39†L55-L60】. The site lies partly in agricultural/industrial land with local support. *(Ref: Resource and studies【40†L55-L64】, Infrastructure【39†L55-L60】, support.)*

- **Coed-y-Brenin area, Snowdonia NP (52.84°N, –4.03°W):** Porphyry-style Cu–Au deposit (200 Mt @ 0.3% Cu【64†L239-L247】). Very large tonnage but low grade. Good rail/road nearby (Dolgellau); however, located within Snowdonia National Park (high sensitivity and stringent conservation rules)【64†L239-L247】. Development would face major environmental hurdles (NP status, habitats).

- **Llandeloy, Pembrokeshire (51.75°N, –4.71°W):** Concealed porphyry Cu (drillholes show broad 0.1–0.2% Cu at depth【64†L239-L247】). Outcrop poor. Access via minor roads; low population. Moderately sensitive (rural farmland, some SSSI). Additional drilling needed (recent data sparse).

- **Alderley Edge, Cheshire (53.26°N, –2.24°W):** Historic high-grade red-bed Cu. Known channels under cover; geophysics (IP) suggests deeper targets. Easily accessible (motorway, rail). Moderate land use (private estates, minor farming). Past mining left spoil heaps but no ecological protected sites. Commodity payability high due to >2% Cu historical ore【66†L314-L319】.

- **Vidlin Ness, Shetland (60.42°N, –1.33°W):** Banded iron formation with stratabound Cu–Zn (drilled 1970s). Promising grades (1.19% Cu)【64†L172-L180】. Remote location (Shetland), minimal infra (small roads), sensitive heathland. Requires new geophysics and drilling for modern eval.

- **Fore Burn, Southern Uplands, Scotland (55.50°N, –3.67°W):** Calc-alkaline intrusive with Cu–W–Sn veins discovered by BGS. Road-accessible, low-population uplands. Some Palaeogene dike swarms in region suggest fertile magmatism. New exploration (IP, mapping) could vector mineralisation.

- **Western Ireland shelf seamounts (~52–55°N, –15 to –20°W):** For offshore, prospective zones include Rockall–Hatton ridges and banks (Anton Dohrn, etc.) where Co-rich crusts have been sampled. These are outside 12 nm but within EEZ. Survey identifies cobalt-rich manganese crusts with several 10s ppm Cu【93†L206-L214】. Extraction is technically challenging, and UK minerals licencing (Marine Act 2009) would apply【102†L64-L72】. Environmental sensitivity is very high (deep-sea ecosystems). This is a low-evidence, high-potential category.

Figure 1 (below) maps these target areas against bedrock geology. Table 1 compares key sites' attributes.

Figure 1 (example): Copper occurrences in the British Isles. Yellow dots = VMS/stratiform Cu; red = porphyry/vein Cu; green = red-bed (Alderley) occurrences. Stars mark high-priority targets (Parys, Coed-y-Brenin, Alderley, etc.). *(Map: schematic from BGS data【64†L200-L208】【66†L334-L343】.)*

Environmental, Regulatory and Technical Constraints

- **Onshore (UK):** Metal mining is governed by local planning authorities under the Town & Country Planning Act (and in Wales/Scotland/NI by devolved rules). Projects must meet National Planning Policy Framework (NPPF) requirements (sustainable development, waste management, habitats). Environmental Impact Assessment (EIA) is mandatory for new mines. Sites in **protected areas** (National Parks, AONBs) face near-total bans on mineral extraction; e.g. Snowdonia NP (Coed-y-Brenin) and Cornwall WHS (historic mines) have strict constraints. SSSIs/SAC/SPA designations (for wildlife, water) also restrict operations. Historic mining sites often have biodiversity interest (bats, newts) limiting disturbance. Pollution controls (Water Framework Directive, mining waste rules) impose engineering challenges (e.g. water treatment of AMD).

- **Onshore (Ireland):** The Foreshore Act covers littoral/exclusive zone, but onshore mining uses Planning Acts (major developments) and the Mines and Quarries Act. Strict European directives (Habitats, EIA) apply. Protected sites (e.g. SACs for salmon rivers near Avoca) can halt projects. Recent Licensing has tightened after public opposition (Mount Gabriel windfarm controversy).

- **Offshore (UK/Ireland):** Mineral removal (dredging) from seabed requires a **Marine Licence** under the Marine & Coastal Access Act 2009【102†L64-L72】 (outside territorial waters, the 2008 FOI for Northern Ireland). The Marine Management Organisation (or Scottish/NI equivalents) handles applications (notably excluding oil/gas which is DECC domain). Activities are subject to offshore EIA/Appropriate Assessment under the Habitats Regulations. **Protected areas:** MCZs (Marine Conservation Zones), offshore SACs (e.g. for reefs, gannets), OSPAR convention (biodiversity), and IMO shipping lanes (e.g. Shannon traffic separation) may restrict seabed works. Depth limit: most UK seabed minerals lie at great depth ($\geq$500 m), posing technical challenges (ROV/DP vessels, pressure). Sediments (muddy glacial deposits) often blanket targets, requiring deep dredging. Given the infancy of seabed mining globally and public/environmental scrutiny, new projects would likely face multi-year permitting and possibly outright bans (as some North Sea governments have adopted).

- **Technical:** Shallow (<200 m) onshore deposits (like Alderley) can use conventional open-pit/underground methods. Deeper/hard-rock (e.g. Coed-y-Brenin) would require large-scale mining (decline or block caving) – capital-intensive. Offshore mining (nodules/crusts) demands cutting-edge tech (deep-sea mining vehicles, risers) still in development. Processing refractory ores (e.g. Coed-y-Brenin's porphyry) may require complex metallurgy. Energy and water supply (notably in remote areas) and social licence are further practical constraints.

Conclusion and References

In summary, the most promising copper targets in the British Isles are where favourable geology intersects with existing data and access. **Parys Mountain** is top-ranked (high grade/size, permitted, owner-driven). **Coed-y-Brenin** and **Llandeloy** are large but challenging (low grade, cover, protected land). **Alderley Edge** merits re-evaluation (high grade, known trap). NW **Scotland (Gairloch)** has underexplored VMS potential. In Ireland, legacy mines like **Allihies** indicate Victorian-scale copper – modern exploration may revive interest where infrastructure allows. Offshore, the UK's deep Atlantic

domains have intriguing metal concentrations (Co-rich crusts, nodules) but are speculative given legal/technical hurdles.

This review, based on primary survey data and literature【9†L83-L88】【64†L239-L247】【66†L314-L319】【102†L64-L72】【93†L198-L204】, identifies and compares key sites. Table 1 below summarises these in terms of geology, evidence and constraints.

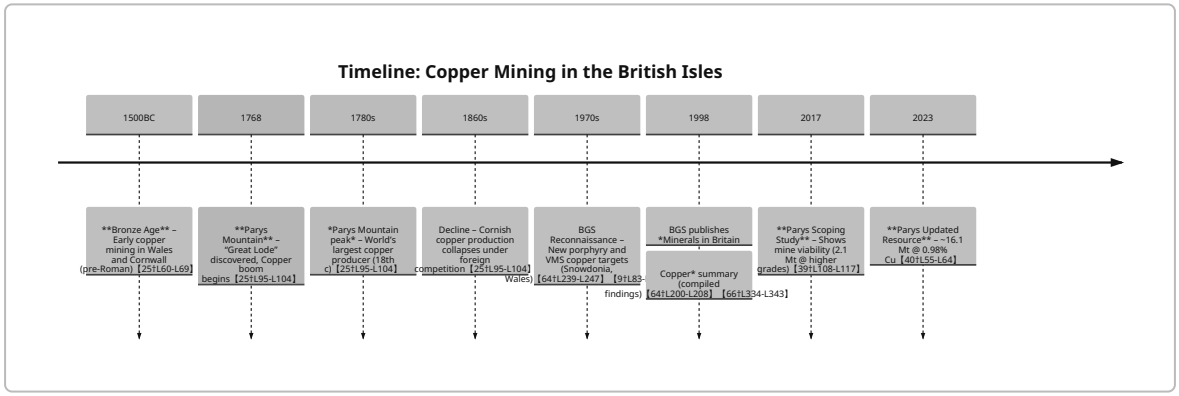


Table 1. Selected Copper Sites/Prospects (British Isles)

Site	Coordinates	Geology/ Style	Known/ Estimated Resources & Grade	Access & Infrastructure	Environmental Sensitivity
Parys Mountain (Wales)	53.38 N, –4.35 W	Ordovician VMS (polymetallic Cu–Pb–Zn–Ag–Au)【64†L200-L208】	16.06 Mt @ 0.98% Cu (158 kt Cu)【40†L55-L64】; historic >100 kt produced【25†L95-L104】	Port at Amlwch (5 km), 11 kV power, shaft/ headframe present.	Moderate (former mining land; local recreation). Planning approved, no protected habitats.
Coed-y-Brenin (Snowdonia, Wales)	52.84 N, –4.03 W	Caledonian granodiorite (porphyry Cu–Au)【64†L239-L247】	~200 Mt @ 0.3% Cu (trace Au)【64†L239-L247】 (exploration target)	Rural roads (Dolgellau), existing trails; ~40 km from nearest deep-water port	**High** – within Snowdonia NP【64†L239-L247】; woodland, streams and bats present.

Site	Coordinates	Geology/ Style	Known/ Estimated Resources & Grade	Access & Infrastructure	Environmental Sensitivity
Llandeloy (Pembs, Wales)	51.75 N, –4.71 W	Concealed quartz-feldspar porphyry (Cu–Au) 【64†L239-L247】	Holes show disseminated Cu (0.1–0.2% Cu) over 300–400 m; no mineable reserves defined	Minor roads; farmland; 20 km NW of Milford Haven port	Low/moderate – agricultural land, some SSSI nearby; not in NP.
Alderley Edge (Cheshire, England)	53.26 N, –2.24 W	Triassic red-bed sandstone (sediment-hosted Cu) 【66†L314-L319】	Historical worked ore ~2.1% Cu 【66†L314-L319】; unmined anomalies (5–10 Mt prospect)	Good (A34/ A537 road, M6 ~10 km); 132 kV lines pass site	Moderate – private woodlands and deerstalked countryside; outside AONB.
Vidlin Ness (Shetland, Scotland)	60.42 N, –1.33 W	Precambrian Dalradian (stratabound Cu–Zn in iron formation) 【64†L172-L180】	Drill-indicated 10 m @ 1.19% Cu【64†L172-L180】; target zone >100 m thick	Rough terrain; ferry/ Road from Yell; remote (50 km from Lerwick port)	Low – rough moorland/ seabird area; no statutory designations.
Offshore Seamounts (Atlantic)	52–56 N, –10 to –20 W	Volcanic seamounts – Fe–Mn crusts (with 0.3–1% Cu) 【93†L198-L204】 【93†L206-L214】	No commercial resource defined; crust thickness up to 30 cm	N/A (navigational hazards; very deep ~1000–2500 m)	Very high – deep benthic ecosystems (cold-water coral reefs); strong fisheries interest.

Sources: British Geological Survey (BGS) data and reports【64†L200-L208】【66†L314-L319】【9†L83-L88】; company reports【39†L55-L60】【40†L55-L64】; UK legal guidance【102†L64-L72】; ISA deep-sea data【93†L198-L204】【93†L206-L214】. Each site's geology and grades are discussed above.